Technical Reproducibility Validation Report

STRspy

Version v2.0
Autosomal STR (ONT)

Verification date	2026-11-01
Repository commit	dafdee7e7e5672c8dc732e8577dbe153f53a12f5
Result	**Does not run as documented: stops at run**
Reached	Installs
Permalink	https://strhub.app/verified/strspy-ont
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Does not run as documented: stops at run**
REACHED	Installs, 2/5 gates passed
WHY	The tool exited with an error before producing its documented output. The installed program starts: it answers --help.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. As it is in the repository: Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	—
Expected IO	Produces a non-empty file in the declared format	—
Plausible Output	Output structure matches expected genotype-bearing format	—
Execution log	strspy-ont.log-external.txt	

Stopping at a step is not a finding that the software is faulty. It records how far this particular attempt got. Software can stop early because a dependency it names is no longer available, because it expects input arranged differently from the reference sample, or because the automated environment cannot supply something it needs, such as commercially licensed components. The messages on screen when it stopped are reported below.

3. What the Author Documents as a Known Issue

Quoted from the repository's README at the verified commit. STRhub did not establish any of this by running the tool; it is the author's own note about their own software.

Known bug (README line 291)

When using the wrapper ('STRspy_run_v1.0.sh'), One might run into an issue. The parallel version of **STRspy_Parallel_v2.0_Args.sh** is unable to properly connect with "gnu parallel" and exits the workflow without mapping or further analysis steps of the pipeline. Solution to this, the user can choose the Normal version to avoid any crash. ***Tip: Its good practice to use pre-aligned bams for quicker outcomes.***

4. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · STRspy v2.0 · commit dafdee7
$ cd '/opt/tool' &&
touch /tmp/.strhub_mark &&
( bash ./STRspy_run_v2.0_Args.sh config/InputConfig.txt config/ToolsConfig.txt ); rc=$?; find . \
-xdev \
-type f \
-newer /tmp/.strhub_mark \
-size +0 \
! \
-path './.git/*' \
! \
-path './proc/*' \
! \
-path './sys/*' \
! \
-path './dev/*' \
! \
-path './data/*' \
2>/dev/null |
head \
-500 |
while IFS= read \
-r f; \
do mkdir \
-p "/data/out/$(dirname \
"$f")" &&
cp "$f" "/data/out/$f"; done; exit $rc
```

5. Reproducibility Metadata

Tool	STRspy
Version	v2.0
Repository	https://github.com/unique379r/strspy
Commit (pinned)	dafdee7e7e5672c8dc732e8577dbe153f53a12f5
Container OS	ubuntu-22.04
Marker panel	Autosomal STR (ONT)
Verified on	2026-11-01 12:48:24 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/41308011428

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

6. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

7. Output Content Evidence

Output file	**/*.t[sx][vt]
Format	TSV
Sequence records	Not assessed
Distinct STR loci	Not assessed
Total reads	Not assessed
Max depth (single locus)	Not assessed

8. Verification Data

The repository ships example data (3 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	1000 Genomes ONT, hg38 CODIS slice (R10 SUP)
Source	https://s3.amazonaws.com/1000g-ont/index.html?prefix=PROCESS…
License	Open access (1000 Genomes / HPRC). Research use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	—

9. Verification Matrix

—	Reference dataset	1000 Genomes ONT, hg38 CODIS slice (R10 SUP)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

10. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.
– A container environment, built from the tool's declared install steps rather than from a recipe the repository ships.

11. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

– Environment: generated by STRhub from the repository's setup/STRspy_2.0_env.yml (conda), at the pinned commit.
– Input: the README documents BAM and FASTQ (line 44); this run used STRhub's ont-bam-hg38 reference data. The author recommends BAM (line 295: "Tip: Its good practice to use pre-aligned bams for quicker outcomes.").
– Run command: the README's own command, rewritten to STRhub's mounts; everything it created was captured as output.

12. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

13. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

14. Conclusion

- **Does not run end-to-end**

  STRspy v2.0 installs in a clean ubuntu-22.04 environment at the pinned commit, but the documented command stopped with an error before finishing. The messages it printed are in the execution log; Section 4 shows how far it got.

- **No output produced**

  No file in the declared format was written, so nothing could be checked for structure or content. This says the run did not get that far; it is not a finding about the quality of the software.

- **Verdict: Fails**

  The tool exited with an error before producing its documented output. The installed program starts: it answers --help.

- **Repository test data not used**

  The repository ships example data (3 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied 1000 Genomes ONT, hg38 CODIS slice (R10 SUP) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

- **Reproducibility statement**

  The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.